

Class 9: Candy Mini-Project

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1. Background

Today we are taking a detour to analyze a fun dataset (that we have more intrinsic insight into) with the most useful analysis method we have learned thus far - Principal Component Analysis (PCA).

2. Data import

The data is all related to halloween candy and is from the 538 website.

```
candy_file <- "candy-data.txt"
candy <- read.csv(candy_file, row.names = 1)
head(candy)
```

	chocolate	fruity	caramel	peanut	almond	nougat	crisped	rice	wafer
100 Grand	1	0	1			0	0		1
3 Musketeers	1	0	0			0	1		0

One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0
	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

There are 85 different types of candy.

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity)
```

```
[1] 38
```

There are 38 fruity candy types in the dataset.

What is your favorite candy?

One of the most interesting variables in the dataset is `winpercent`. For a given candy this value is the percentage of people who prefer this candy over another randomly chosen candy from the dataset (what 538 term a matchup). Higher values indicate a more popular candy.

For example, with Twix:

```
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

An alternate way is with the `dplyr` package.

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
candy |>
  filter(row.names(candy)=="Twix") |>
  select(winpercent)
```

```
      winpercent
Twix    81.64291
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```
library(dplyr)

candy |>
  filter(row.names(candy)=="Sour Patch Kids") |>
  select(winpercent)
```

```
      winpercent
Sour Patch Kids    59.864
```

My favorite candy is Sour Patch Kids, and the winpercent value is 59.864.

Q4. What is the winpercent value for "Kit Kat"?

```
library(dplyr)

candy |>
  filter(row.names(candy)=="Kit Kat") |>
  select(winpercent)
```

```

      winpercent
Kit Kat      76.7686

```

The winpercent value for Kit Kat is 76.7686.

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```

library(dplyr)

candy |>
  filter(row.names(candy)=="Tootsie Roll Snack Bars") |>
  select(winpercent)

```

```

      winpercent
Tootsie Roll Snack Bars  49.6535

```

The winpercent value for Tootsie Roll Snack Bars is 49.6535.

N.B: There is a useful `skim()` function in the **skimr** package that can help give you a quick overview of a given dataset. Let’s install this package and try it on our candy data.

```

library("skimr")
skim(candy)

```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

A shortcut:

```
skimr::skim(candy)
```

Table 3: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Yes, the column “winpercent” stands out because most of the variables (chocolate, fruity, caramel, etc.) are 0 or 1, but winpercent has values ranging from 0-100.

Q7. What do you think a zero and one represent for the `candy$chocolate` column?

```
table(candy$chocolate)
```

```
0 1
48 37
```

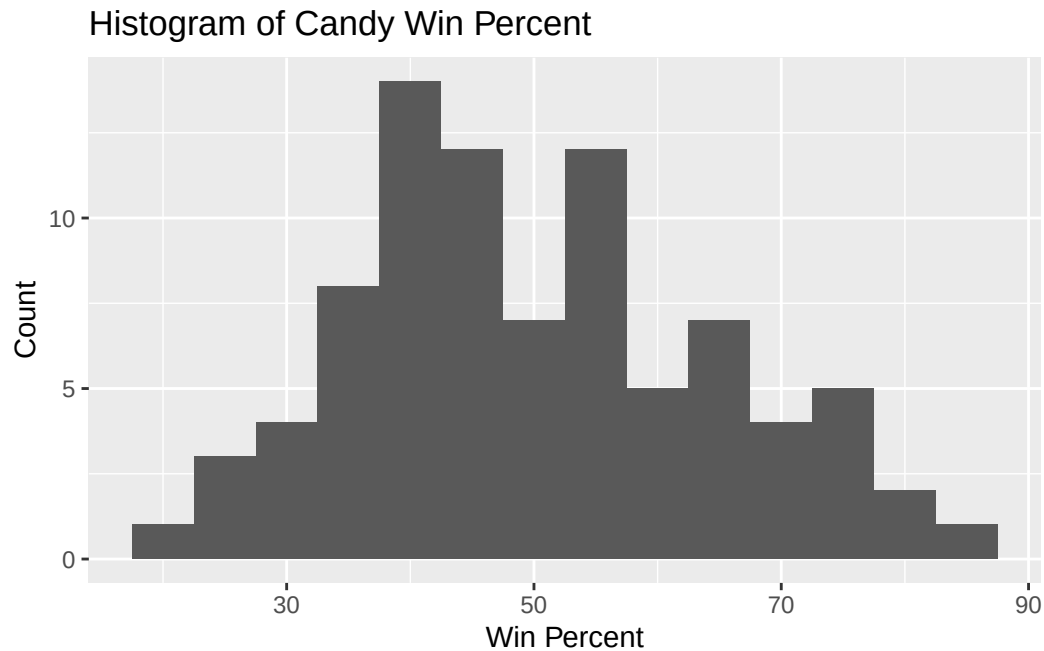
The meaning of 0 and 1 in `candy$chocolate` is that 1= the candy has chocolate and 0= the candy does not have chocolate.

3. Exploratory Analysis

Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```
library(ggplot2)

ggplot(candy, aes(x = winpercent)) +
  geom_histogram(binwidth = 5) +
  labs(title = "Histogram of Candy Win Percent",
       x = "Win Percent",
       y = "Count")
```



Q9. Is the distribution of winpercent values symmetrical?

The distribution of winpercent values is not symmetrical, it is slightly right-skewed with the lower values.

Q10. Is the center of the distribution above or below 50%?

The center of distribution is slightly above 50%.

```
mean(candy$winpercent)
```

```
[1] 50.31676
```

```
summary(candy$winpercent)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22.45	39.14	47.83	50.32	59.86	84.18

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
# Line by line for chocolate candy
choc.ind <- as.logical(candy$chocolate)
choc.candy <- candy[choc.ind, ]
choc.win <- choc.candy$winpercent
mean(choc.win)
```

```
[1] 60.92153
```

```
# Line by line for fruity candy
fruit.ind <- as.logical(candy$fruity)
fruit.candy <- candy[fruit.ind, ]
fruit.win <- fruit.candy$winpercent
mean(fruit.win)
```

```
[1] 44.11974
```

```
# Both chocolate and fruity candy in 1 code
mean(candy$winpercent[as.logical(candy$chocolate)])
```

```
[1] 60.92153
```

```
mean(candy$winpercent[as.logical(candy$fruity)])
```

```
[1] 44.11974
```

Chocolate candy is higher on average than fruit candy: $60.92153 > 44.11974$.

Q12. Is this difference statistically significant?

```
t.test(candy$winpercent[as.logical(candy$chocolate)],
       candy$winpercent[as.logical(candy$fruity)])
```

Welch Two Sample t-test

```
data: candy$winpercent[as.logical(candy$chocolate)] and candy$winpercent[as.logical(candy$fruity)]
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
```



```
11.44563 22.15795
sample estimates:
mean of x mean of y
60.92153 44.11974
```

This difference is statistically different because the p.value 2.871e-08 is smaller than 0.05.

4. Overall Candy Rankings

Let's use the base R `order()` function together with `head()` to sort the whole dataset by a variable/column of interest. Or if you have been getting into the tidyverse and the dplyr package you can use the `arrange()` function together with `head()` to do the same thing and answer the following questions:

Q13. What are the five least liked candy types in this set?

```
candy |>
  arrange(winpercent) |>
  select(winpercent) |>
  head(5)
```

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

The five least liked candy types in this set are: 1. Nik L Nip, 2. Boston Baked Beans, 3. Chiclets, 4. Super Bubble, and 5. Jawbusters.

```
x <-c(5,10,1)
order(x)
```

```
[1] 3 1 2
```

```
x[ order (x) ]
```

```
[1] 1 5 10
```

```
candy [ order(candy$winpercent),]
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Nik L Nip	0	1	0	0	0
Boston Baked Beans	0	0	0	1	0
Chiclets	0	1	0	0	0
Super Bubble	0	1	0	0	0
Jawbusters	0	1	0	0	0
Root Beer Barrels	0	0	0	0	0
Sugar Daddy	0	0	1	0	0
One dime	0	0	0	0	0
Sugar Babies	0	0	1	0	0
Haribo Happy Cola	0	0	0	0	0
Caramel Apple Pops	0	1	1	0	0
Strawberry bon bons	0	1	0	0	0
Sixlets	1	0	0	0	0
Ring pop	0	1	0	0	0
Chewey Lemonhead Fruit Mix	0	1	0	0	0
Red vines	0	1	0	0	0
Pixie Sticks	0	0	0	0	0
Nestle Smarties	1	0	0	0	0
Candy Corn	0	0	0	0	0
Charleston Chew	1	0	0	0	1
Warheads	0	1	0	0	0
Lemonhead	0	1	0	0	0
Fun Dip	0	1	0	0	0
Now & Later	0	1	0	0	0
Dum Dums	0	1	0	0	0
Pop Rocks	0	1	0	0	0
Laffy Taffy	0	1	0	0	0
Werther's Original Caramel	0	0	1	0	0
Haribo Twin Snakes	0	1	0	0	0
Dots	0	1	0	0	0
Runts	0	1	0	0	0
Tootsie Roll Juniors	1	0	0	0	0
Fruit Chews	0	1	0	0	0
Welch's Fruit Snacks	0	1	0	0	0
Twizzlers	0	1	0	0	0
Tootsie Roll Midgies	1	0	0	0	0
Smarties candy	0	1	0	0	0
One quarter	0	0	0	0	0
Payday	0	0	0	1	1

Mike & Ike	0	1	0	0	0
Gobstopper	0	1	0	0	0
Trolli Sour Bites	0	1	0	0	0
Mounds	1	0	0	0	0
Tootsie Pop	1	1	0	0	0
Whoppers	1	0	0	0	0
Tootsie Roll Snack Bars	1	0	0	0	0
Almond Joy	1	0	0	1	0
Haribo Sour Bears	0	1	0	0	0
Air Heads	0	1	0	0	0
Sour Patch Tricksters	0	1	0	0	0
Lifesavers big ring gummies	0	1	0	0	0
Mr Good Bar	1	0	0	1	0
Swedish Fish	0	1	0	0	0
Milk Duds	1	0	1	0	0
Skittles wildberry	0	1	0	0	0
Nerds	0	1	0	0	0
Hershey's Kisses	1	0	0	0	0
Hershey's Milk Chocolate	1	0	0	0	0
Baby Ruth	1	0	1	1	1
Haribo Gold Bears	0	1	0	0	0
Junior Mints	1	0	0	0	0
Hershey's Special Dark	1	0	0	0	0
Snickers Crisper	1	0	1	1	0
Sour Patch Kids	0	1	0	0	0
Milky Way Midnight	1	0	1	0	1
Hershey's Krackel	1	0	0	0	0
Skittles original	0	1	0	0	0
Milky Way Simply Caramel	1	0	1	0	0
Rolo	1	0	1	0	0
Nestle Crunch	1	0	0	0	0
M&M's	1	0	0	0	0
100 Grand	1	0	1	0	0
Starburst	0	1	0	0	0
3 Musketeers	1	0	0	0	1
Peanut M&Ms	1	0	0	1	0
Nestle Butterfinger	1	0	0	1	0
Peanut butter M&M's	1	0	0	1	0
Reese's stuffed with pieces	1	0	0	1	0
Milky Way	1	0	1	0	1
Reese's pieces	1	0	0	1	0
Snickers	1	0	1	1	1
Kit Kat	1	0	0	0	0

Twix	1	0	1	0	0
Reese's Miniatures	1	0	0	1	0
Reese's Peanut Butter cup	1	0	0	1	0
	crisped	ricewafer	hard bar	pluribus	sugarpercent
Nik L Nip	0	0	0	1	0.197
Boston Baked Beans	0	0	0	1	0.313
Chiclets	0	0	0	1	0.046
Super Bubble	0	0	0	0	0.162
Jawbusters	0	1	0	1	0.093
Root Beer Barrels	0	1	0	1	0.732
Sugar Daddy	0	0	0	0	0.418
One dime	0	0	0	0	0.011
Sugar Babies	0	0	0	1	0.965
Haribo Happy Cola	0	0	0	1	0.465
Caramel Apple Pops	0	0	0	0	0.604
Strawberry bon bons	0	1	0	1	0.569
Sixlets	0	0	0	1	0.220
Ring pop	0	1	0	0	0.732
Chewey Lemonhead Fruit Mix	0	0	0	1	0.732
Red vines	0	0	0	1	0.581
Pixie Sticks	0	0	0	1	0.093
Nestle Smarties	0	0	0	1	0.267
Candy Corn	0	0	0	1	0.906
Charleston Chew	0	0	1	0	0.604
Warheads	0	1	0	0	0.093
Lemonhead	0	1	0	0	0.046
Fun Dip	0	1	0	0	0.732
Now & Later	0	0	0	1	0.220
Dum Dums	0	1	0	0	0.732
Pop Rocks	0	1	0	1	0.604
Laffy Taffy	0	0	0	0	0.220
Werther's Original Caramel	0	1	0	0	0.186
Haribo Twin Snakes	0	0	0	1	0.465
Dots	0	0	0	1	0.732
Runts	0	1	0	1	0.872
Tootsie Roll Juniors	0	0	0	0	0.313
Fruit Chews	0	0	0	1	0.127
Welch's Fruit Snacks	0	0	0	1	0.313
Twizzlers	0	0	0	0	0.220
Tootsie Roll Midgies	0	0	0	1	0.174
Smarties candy	0	1	0	1	0.267
One quarter	0	0	0	0	0.011
Payday	0	0	1	0	0.465

Mike & Ike	0	0	0	1	0.872
Gobstopper	0	1	0	1	0.906
Trolli Sour Bites	0	0	0	1	0.313
Mounds	0	0	1	0	0.313
Tootsie Pop	0	1	0	0	0.604
Whoppers	1	0	0	1	0.872
Tootsie Roll Snack Bars	0	0	1	0	0.465
Almond Joy	0	0	1	0	0.465
Haribo Sour Bears	0	0	0	1	0.465
Air Heads	0	0	0	0	0.906
Sour Patch Tricksters	0	0	0	1	0.069
Lifesavers big ring gummies	0	0	0	0	0.267
Mr Good Bar	0	0	1	0	0.313
Swedish Fish	0	0	0	1	0.604
Milk Duds	0	0	0	1	0.302
Skittles wildberry	0	0	0	1	0.941
Nerds	0	1	0	1	0.848
Hershey's Kisses	0	0	0	1	0.127
Hershey's Milk Chocolate	0	0	1	0	0.430
Baby Ruth	0	0	1	0	0.604
Haribo Gold Bears	0	0	0	1	0.465
Junior Mints	0	0	0	1	0.197
Hershey's Special Dark	0	0	1	0	0.430
Snickers Crisper	1	0	1	0	0.604
Sour Patch Kids	0	0	0	1	0.069
Milky Way Midnight	0	0	1	0	0.732
Hershey's Krackel	1	0	1	0	0.430
Skittles original	0	0	0	1	0.941
Milky Way Simply Caramel	0	0	1	0	0.965
Rolo	0	0	0	1	0.860
Nestle Crunch	1	0	1	0	0.313
M&M's	0	0	0	1	0.825
100 Grand	1	0	1	0	0.732
Starburst	0	0	0	1	0.151
3 Musketeers	0	0	1	0	0.604
Peanut M&Ms	0	0	0	1	0.593
Nestle Butterfinger	0	0	1	0	0.604
Peanut butter M&M's	0	0	0	1	0.825
Reese's stuffed with pieces	0	0	0	0	0.988
Milky Way	0	0	1	0	0.604
Reese's pieces	0	0	0	1	0.406
Snickers	0	0	1	0	0.546
Kit Kat	1	0	1	0	0.313

Twix	1	0	1	0	0.546
Reese's Miniatures	0	0	0	0	0.034
Reese's Peanut Butter cup	0	0	0	0	0.720

	price	percent	win	percent	
Nik L Nip	0.976	22.44	534		
Boston Baked Beans	0.511	23.41	782		
Chiclets	0.325	24.52	499		
Super Bubble	0.116	27.30	386		
Jawbusters	0.511	28.12	744		
Root Beer Barrels	0.069	29.70	369		
Sugar Daddy	0.325	32.23	100		
One dime	0.116	32.26	109		
Sugar Babies	0.767	33.43	755		
Haribo Happy Cola	0.465	34.15	896		
Caramel Apple Pops	0.325	34.51	768		
Strawberry bon bons	0.058	34.57	899		
Sixlets	0.081	34.72	200		
Ring pop	0.965	35.29	076		
Chewey Lemonhead Fruit Mix	0.511	36.01	763		
Red vines	0.116	37.34	852		
Pixie Sticks	0.023	37.72	234		
Nestle Smarties	0.976	37.88	719		
Candy Corn	0.325	38.01	096		
Charleston Chew	0.511	38.97	504		
Warheads	0.116	39.01	190		
Lemonhead	0.104	39.14	106		
Fun Dip	0.325	39.18	550		
Now & Later	0.325	39.44	680		
Dum Dums	0.034	39.46	056		
Pop Rocks	0.837	41.26	551		
Laffy Taffy	0.116	41.38	956		
Werther's Original Caramel	0.267	41.90	431		
Haribo Twin Snakes	0.465	42.17	877		
Dots	0.511	42.27	208		
Runts	0.279	42.84	914		
Tootsie Roll Juniors	0.511	43.06	890		
Fruit Chews	0.034	43.08	892		
Welch's Fruit Snacks	0.313	44.37	552		
Twizzlers	0.116	45.46	628		
Tootsie Roll Midgies	0.011	45.73	675		
Smarties candy	0.116	45.99	583		
One quarter	0.511	46.11	650		
Payday	0.767	46.29	660		

Mike & Ike	0.325	46.41172
Gobstopper	0.453	46.78335
Trolli Sour Bites	0.255	47.17323
Mounds	0.860	47.82975
Tootsie Pop	0.325	48.98265
Whoppers	0.848	49.52411
Tootsie Roll Snack Bars	0.325	49.65350
Almond Joy	0.767	50.34755
Haribo Sour Bears	0.465	51.41243
Air Heads	0.511	52.34146
Sour Patch Tricksters	0.116	52.82595
Lifesavers big ring gummies	0.279	52.91139
Mr Good Bar	0.918	54.52645
Swedish Fish	0.755	54.86111
Milk Duds	0.511	55.06407
Skittles wildberry	0.220	55.10370
Nerds	0.325	55.35405
Hershey's Kisses	0.093	55.37545
Hershey's Milk Chocolate	0.918	56.49050
Baby Ruth	0.767	56.91455
Haribo Gold Bears	0.465	57.11974
Junior Mints	0.511	57.21925
Hershey's Special Dark	0.918	59.23612
Snickers Crisper	0.651	59.52925
Sour Patch Kids	0.116	59.86400
Milky Way Midnight	0.441	60.80070
Hershey's Krackel	0.918	62.28448
Skittles original	0.220	63.08514
Milky Way Simply Caramel	0.860	64.35334
Rolo	0.860	65.71629
Nestle Crunch	0.767	66.47068
M&M's	0.651	66.57458
100 Grand	0.860	66.97173
Starburst	0.220	67.03763
3 Musketeers	0.511	67.60294
Peanut M&Ms	0.651	69.48379
Nestle Butterfinger	0.767	70.73564
Peanut butter M&M's	0.651	71.46505
Reese's stuffed with pieces	0.651	72.88790
Milky Way	0.651	73.09956
Reese's pieces	0.651	73.43499
Snickers	0.651	76.67378
Kit Kat	0.511	76.76860

Twix	0.906	81.64291
Reese's Miniatures	0.279	81.86626
Reese's Peanut Butter cup	0.651	84.18029

Q14. What are the top 5 all time favorite candy types out of this set?

```
candy |>
  arrange(-winpercent) |>
  head(5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1

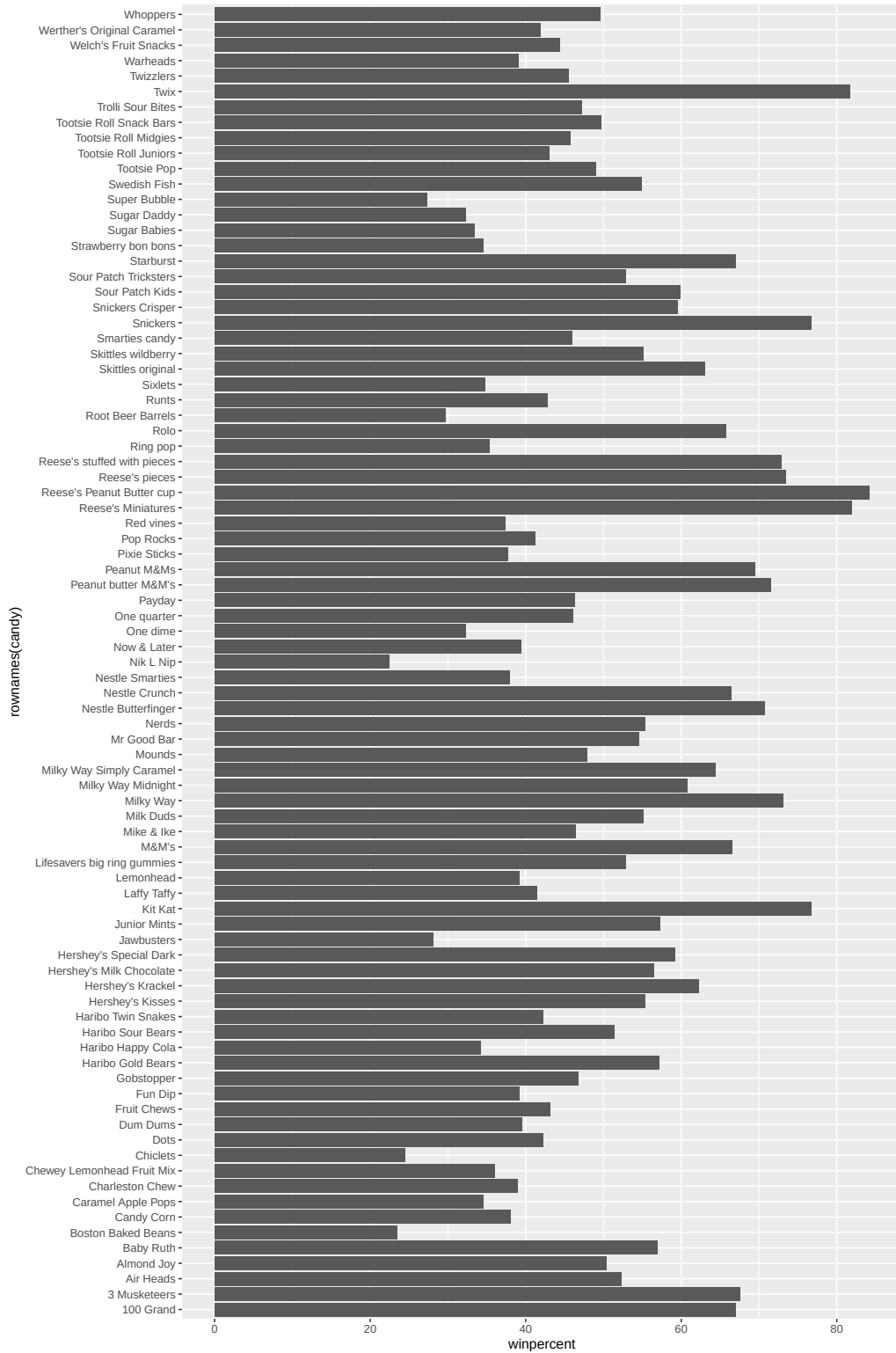
	crisp	rice	waffer	hard	bar	pluribus	sugar
Reese's Peanut Butter cup		0	0	0		0	0.720
Reese's Miniatures		0	0	0		0	0.034
Twix		1	0	1		0	0.546
Kit Kat		1	0	1		0	0.313
Snickers		0	0	1		0	0.546

	price	percent	winpercent
Reese's Peanut Butter cup	0.651	84.18029	
Reese's Miniatures	0.279	81.86626	
Twix	0.906	81.64291	
Kit Kat	0.511	76.76860	
Snickers	0.651	76.67378	

The five most liked candy types in this set are: 1. Reese's Peanut Butter cup, 2. Reese's Miniatures, 3. Twix, 4. Kit Kat, and 5. Snickers.

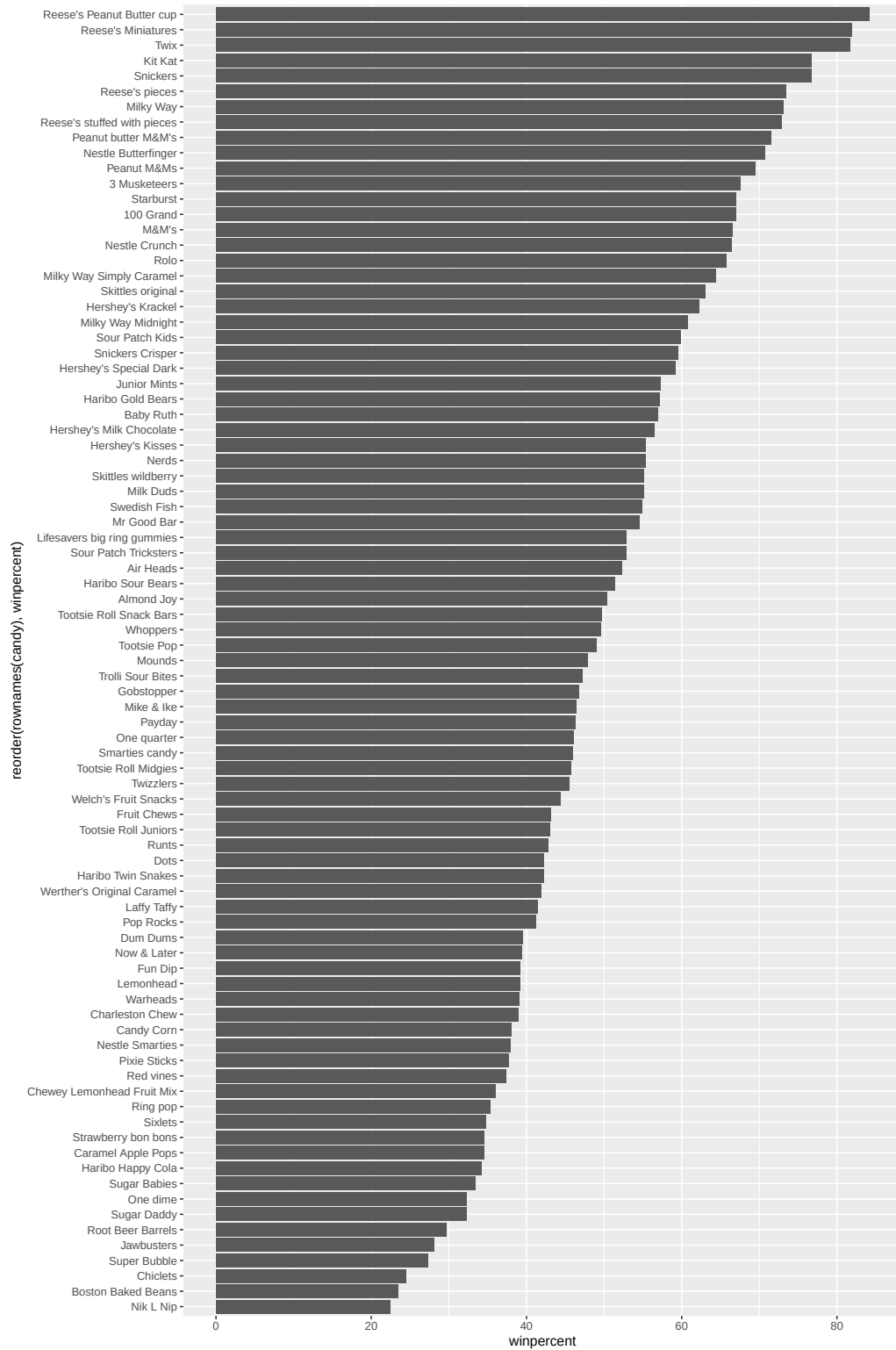
Q15. Make a first barplot of candy ranking based on winpercent values.

```
ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()
```

Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

```
ggplot(candy) +  
  aes(winpercent, reorder(rownames(candy), winpercent)) +  
  geom_col()
```



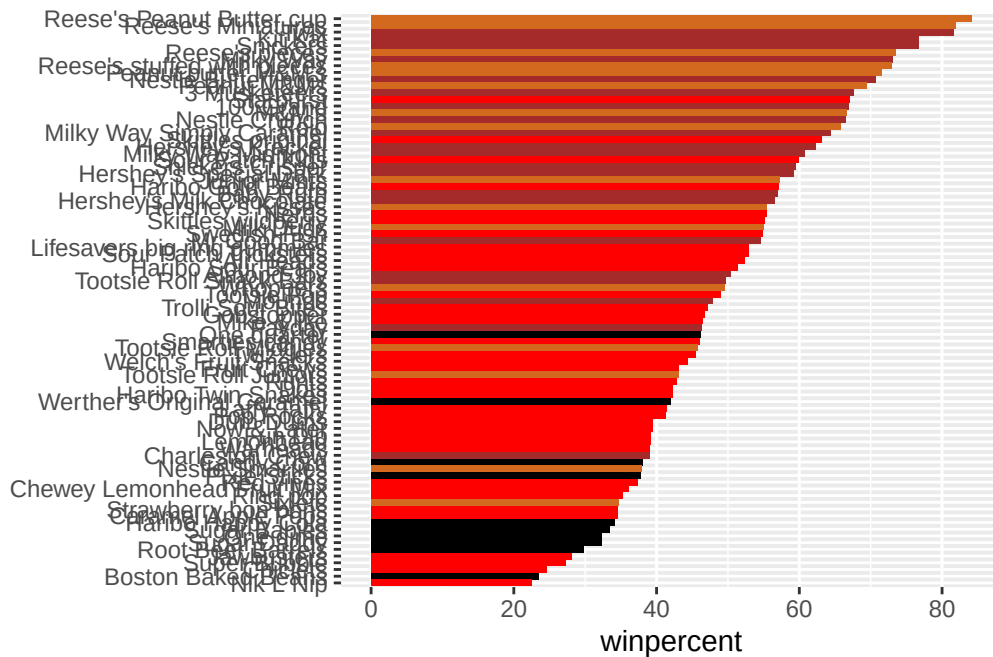
Adding useful color

We start by making a vector of all black values (one for each candy). Then we overwrite chocolate (for chocolate candy), brown (for candy bars) and red (for fruity candy) values.

```
my_cols=rep("black", nrow(candy))

## Chocolate candy
my_cols[as.logical(candy$chocolate)] <- "chocolate"
## Candy bars
my_cols[as.logical(candy$bar)] <- "brown"
## Fruity candy
my_cols[as.logical(candy$fruity)] <- "red"
```

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col(fill=my_cols) +
  ylab("")
```



Q17. What is the worst ranked chocolate candy?

```
choc <- candy[as.logical(candy$chocolate), ]
choc[which.min(choc$winpercent), ]
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer	hard
Sixlets	1	0	0	0	0	0	0

	bar	pluribus	sugarpercent	pricepercent	winpercent
Sixlets	0	1	0.22	0.081	34.722

The worst ranked chocolate candy is Sixlets.

Q18. What is the best ranked fruity candy?

```
fruity <- candy[as.logical(candy$fruity), ]
fruity[which.max(fruity$winpercent), ]
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer	hard
Starburst	0	1	0	0	0	0	0

	bar	pluribus	sugarpercent	pricepercent	winpercent
Starburst	0	1	0.151	0.22	67.03763

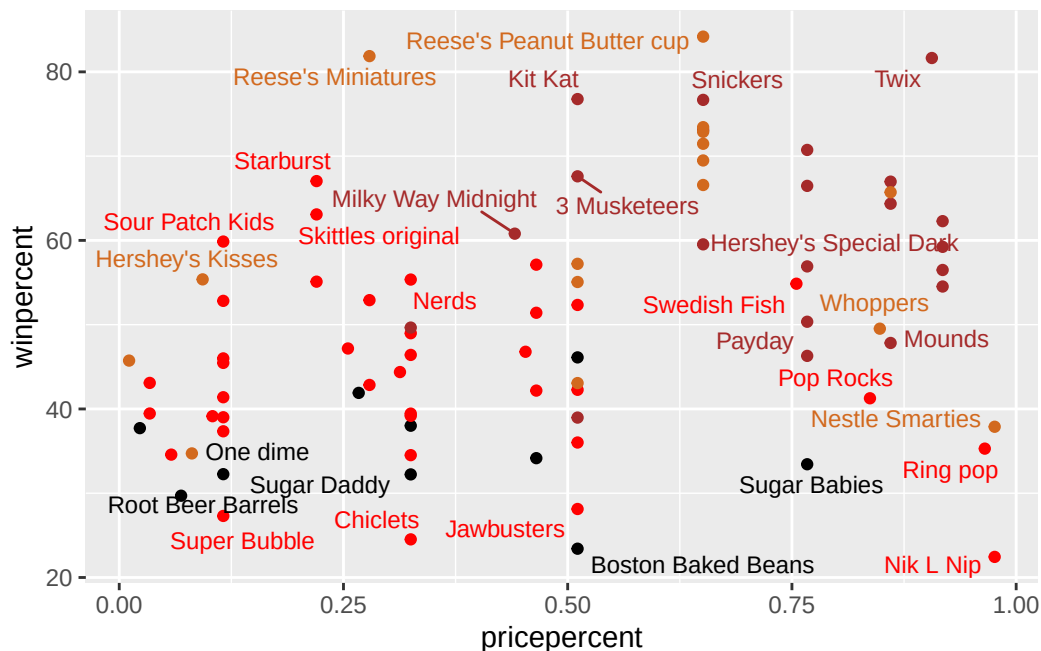
The best ranked fruity candy is Starburst.

5. Taking a look at pricepercent

Make a plot of winpercent vs pricepercent

```
library(ggrepel)

ggplot(candy) +
  aes(x = pricepercent, y = winpercent, label = rownames(candy)) +
  geom_point(col = my_cols) +
  geom_text_repel(col = my_cols, size = 3.3, max.overlaps = 5)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

```
candy[which.max(candy$winpercent / candy$pricepercent), ]
```

```

               chocolate fruity caramel peanuty almondy nougat
Tootsie Roll Midgies           1      0      0           0      0
               crispedricewafer hard bar pluribus sugarpercent
Tootsie Roll Midgies           0      0      0           1      0.174
               pricepercent winpercent
Tootsie Roll Midgies      0.011  45.73675

```

Tootsie Roll Snack Bars is the highest ranked in terms of winpercent for the least money.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```

ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )

```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

```
library(dplyr)
candy |>
  arrange(-pricepercent) |>
  select(pricepercent, winpercent) |>
  head(n=5)
```

	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

The top 5 most expensive candy types are: 1. Nik L Nips, 2. Nestle Smarties, 3. Ring pops, 4. Hershey's Krackel's, and 5. Hershey's Milk Chocolates. Among all of these, the least popular is Nik L Nips because it has the lowest winpercent (22.44534).

6. Exploring the correlation structure

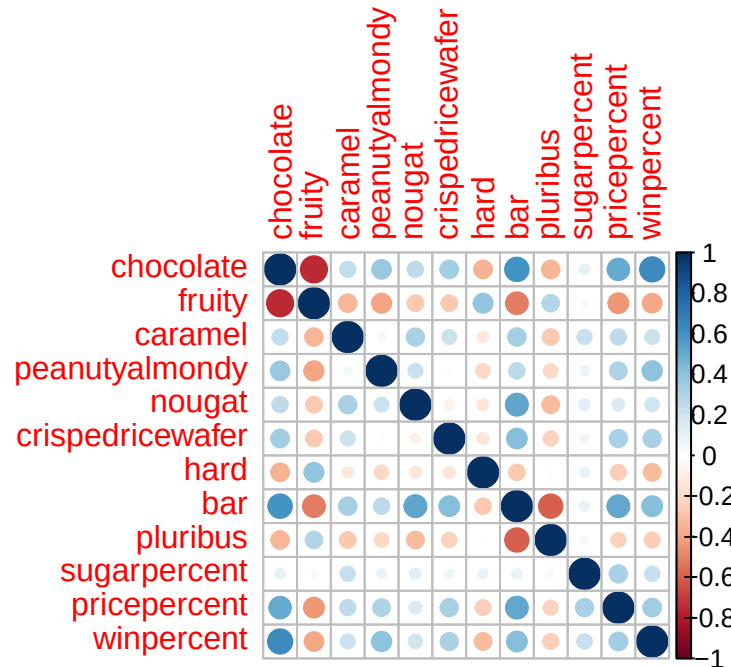
We can calculate the pair-wise correlation of all our columns

N.B: This matters especially because PCA operates not on raw data but on the relationships between variables. Specifically, it finds the directions that capture the most co-variation in your data. This is also why we will use `scale=TRUE` when calling `prcomp()` in the next section: it ensures all variables contribute equally by working from correlations (standardized and scale-free) rather than raw covariances, which would be dominated by whichever variable happens to have the largest numeric range. Can you guess which variable in our dataset that would be? An important point here is to think of the analysis below as a preview of your upcoming PCA results: the patterns we spot here should reappear in the loadings plot at the end of the next section.

```
library(corrplot)
```

corrplot 0.95 loaded

```
cij <- cor(candy)
corrplot(cij)
```



A few things to notice about this correlation matrix plot. First, the diagonal is always perfectly blue (correlation = 1.0) — every variable is perfectly correlated with itself. Second, the matrix is symmetric: the correlation between **chocolate** and **bar** is the same as between **bar** and **chocolate**, so the upper and lower triangles are mirror images. Most importantly, this matrix is exactly what PCA will decompose mathematically. When we run `prcomp()` on scaled data, it is performing an eigen-decomposition of this correlation matrix — finding the directions (eigen-vectors) that explain the most variance (eigen-values). The loading values you will see shortly are those eigen-vectors, and the proportion of variance explained by each PC corresponds to its eigen-value. The corrplot is therefore not just an exploratory tool, it is the structure that PCA will summarize.

Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

The two variables that are anti-correlated are chocolate and fruity. This occurs because candies are either chocolatey or fruity.

Q23. Use your corrplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

Variables that would have the largest contributions to PC1: chocolate and bar (positively) and fruity (negatively).

Principal Component Analysis

In this case we want to be sure to set `scale=TRUE` argument for `prcomp()` because we have one variable `winpercent` that is on a very different scale than all others and would otherwise dominate our PCA results.

```
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369
	PC8	PC9	PC10	PC11	PC12		
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760		
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317		
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000		

First major result figure is the “score plt” of PC1 vs PC2 - how the different candy are related to each other on our new PC axis:

```
pca$x
```

	PC1	PC2	PC3	PC4
100 Grand	-3.81986175	-0.5935787670	-2.186308676	-2.37159574
3 Musketeers	-2.79602364	-1.5196062111	1.412198551	0.69943868
One dime	1.20258363	0.1718120657	2.060771178	-1.20067824
One quarter	0.44865378	0.4519735621	1.476492844	-1.00177141
Air Heads	0.70289922	-0.5731343263	-0.929389343	0.41245656
Almond Joy	-2.46833834	0.7035501120	0.858108916	0.57249739
Baby Ruth	-4.10531223	-2.1000967736	1.347834706	1.85505225
Boston Baked Beans	0.71385813	1.2098216537	0.941899950	1.10219913
Candy Corn	1.01357204	0.2834319621	-0.840681586	0.83664703

Caramel Apple Pops	0.81049645	-1.6960889498	-0.207020586	-0.30186567
Charleston Chew	-2.15436587	-1.9304213037	1.675469334	0.46999498
Chewey Lemonhead Fruit Mix	1.65268482	0.0726434944	-0.909617411	0.58609915
Chiclets	2.38180817	0.4430926071	1.000422079	-0.59998577
Dots	1.51249936	0.1623958592	-0.967135199	0.63622661
Dum Dums	2.14430933	-1.8388386160	-0.385372660	-0.14796280
Fruit Chews	2.26133763	0.5818322520	0.978626618	-0.39164187
Fun Dip	1.82383348	-1.7828662094	-0.719415821	-0.08544003
Gobstopper	1.96047812	-1.0584680267	-1.873874385	0.84237208
Haribo Gold Bears	1.33360746	0.5892699921	-0.431929774	0.33530766
Haribo Happy Cola	1.11167365	0.6257697808	0.054459647	0.16024129
Haribo Sour Bears	1.46152952	0.5073691482	-0.379443632	0.28956535
Haribo Twin Snakes	1.66849016	0.3748646265	-0.294528131	0.21556045
Hershey's Kisses	0.37722675	1.5654519145	1.104739528	-0.30451907
Hershey's Krackel	-3.04788356	0.6850792787	-1.154357778	-2.79294516
Hershey's Milk Chocolate	-2.11696417	0.2504568891	0.218316614	-0.64942872
Hershey's Special Dark	-2.17850376	0.2898570052	0.193067056	-0.62742342
Jawbusters	2.62491587	-0.6343671618	0.114043053	-0.54172092
Junior Mints	-0.16010610	1.6194428347	0.442156347	-0.08935729
Kit Kat	-2.87086546	0.9069655335	-0.545771148	-2.94691419
Laffy Taffy	1.65450042	-0.2379605144	1.217408326	-0.81578254
Lemonhead	2.33564695	-1.2553404646	1.125823900	-1.18755633
Lifesavers big ring gummies	1.19528766	-0.0783610246	0.814040659	-0.61506538
Peanut butter M&M's	-1.52223814	1.9291395890	-0.815897653	2.27060871
M&M's	-0.76747561	1.2573539136	-1.260658369	0.98037043
Mike & Ike	1.57487290	0.0664259746	-1.114406454	0.84284942
Milk Duds	-0.76836937	0.4192793946	-0.137573021	-0.01222490
Milky Way	-3.69272218	-2.4933313173	0.843423990	0.70792542
Milky Way Midnight	-3.23036513	-2.8201031327	0.902884388	0.75904764
Milky Way Simply Caramel	-3.04936226	-1.1774777304	-1.382617058	0.15488883
Mounds	-1.81292795	0.2120726312	0.636094539	-0.91126847
Mr Good Bar	-2.67327849	0.9217207344	0.997161433	0.40634715
Nerds	1.93426895	-0.9133307225	-1.670281710	0.79359684
Nestle Butterfinger	-2.97855081	0.8798835368	0.348599786	0.94918411
Nestle Crunch	-2.92740488	0.8119013154	-0.747159803	-2.97250665
Nik L Nip	1.63985272	0.4210217322	-0.083217936	-0.24015596
Now & Later	1.98070982	0.5117150919	0.460099768	-0.21340533
Payday	-2.39180556	-1.4839637512	2.091687409	1.65680787
Peanut M&Ms	-1.38897069	2.0947188031	-0.260214925	1.89874716
Pixie Sticks	1.67042227	0.8969792365	1.394703254	-0.48031281
Pop Rocks	1.76879348	-0.8060325640	-1.567639814	0.42017068
Red vines	2.12406849	0.1366822960	-0.115183020	0.27721045
Reese's Miniatures	-1.55210251	1.9287569793	1.884620322	0.48605155

Reese's Peanut Butter cup	-2.28427985	1.4648923293	-0.156138940	1.63994489
Reese's pieces	-1.40590761	2.3077984818	0.136661895	1.64348077
Reese's stuffed with pieces	-2.13382398	1.0787289654	-0.673152403	1.96066081
Ring pop	1.19274412	-1.7069749284	-1.423826969	0.02570015
Rolo	-1.61259322	0.1773734932	-1.931879747	1.00697641
Root Beer Barrels	2.10440254	-0.8711340556	-0.594335570	0.36204105
Runts	2.25699185	-1.1223199934	-1.557678507	0.71996790
Sixlets	0.81799664	1.1888290122	1.093105891	-0.33001942
Skittles original	1.29259129	0.2263705137	-1.306145308	1.05900061
Skittles wildberry	1.47148517	0.1118354559	-1.232745536	0.99503181
Nestle Smarties	-0.27556563	1.3792344137	-0.080047831	-0.03345921
Smarties candy	2.60115214	-0.6047947520	0.003482896	-0.21938457
Snickers	-4.39576792	-1.7919312516	1.434654778	1.89861891
Snickers Crisper	-4.01457335	-0.0347673522	-1.089868643	-1.42309414
Sour Patch Kids	1.81551769	0.8879445215	0.863881832	-0.32795118
Sour Patch Tricksters	1.97326660	0.7869473239	0.928605869	-0.38435897
Starburst	1.50658493	0.9437290830	0.487658690	-0.12150268
Strawberry bon bons	2.80647837	-1.0331193111	-0.524069119	0.13960292
Sugar Babies	-0.01900559	-0.8219542293	-1.802826526	0.92212945
Sugar Daddy	0.19642038	-1.2073694698	0.520140143	-0.59663517
Super Bubble	1.99242820	-0.3915898648	1.481310204	-1.01767057
Swedish Fish	1.00547407	0.5003327040	-1.068588828	0.59499272
Tootsie Pop	0.84734171	-1.1060686710	-0.480874078	-0.23675349
Tootsie Roll Juniors	-0.40463667	0.5848580362	0.836999949	-0.59623669
Tootsie Roll Midgies	0.66730732	1.3709464980	1.179339290	-0.32789249
Tootsie Roll Snack Bars	-1.31149842	0.0009721286	0.885976952	-0.78242207
Trolli Sour Bites	1.85048456	0.5304055168	0.254559391	-0.02435072
Twix	-4.12909044	-0.2180299573	-1.943536689	-2.52917855
Twizzlers	1.56312584	-0.1794588354	1.179917535	-0.78310886
Warheads	2.30707033	-1.2940268825	1.004249910	-1.11380521
Welch's Fruit Snacks	1.84808801	0.5022006184	0.213204782	-0.03387254
Werther's Original Caramel	0.68420363	-2.0146385440	0.506488679	-0.97208672
Whoppers	-1.42549552	1.3654147702	-2.759982292	-1.23030133
	PC5	PC6	PC7	PC8
100 Grand	-0.66236243	-0.545218405	-0.14340559	0.577224229
3 Musketeers	-0.16006665	0.382588420	2.02155526	-1.502574960
One dime	-0.26977985	0.094950530	-0.77220066	0.155622104
One quarter	-0.05093737	0.428354037	-0.59089199	0.359134372
Air Heads	0.33108524	0.188791601	0.96808076	1.012393252
Almond Joy	1.43200435	1.022273480	-0.66604596	0.608261288
Baby Ruth	-0.53157179	0.026152894	-0.27477854	0.617961302
Boston Baked Beans	-0.01286555	1.147068667	-1.66290853	1.017203801
Candy Corn	-0.92992691	0.482570575	-0.17268382	-0.433778134

Caramel Apple Pops	-1.26916612	-0.641471733	-0.77030933	1.705704289
Charleston Chew	-0.48839246	1.490541025	1.23825892	-1.445444979
Chewey Lemonhead Fruit Mix	-0.64216946	0.867070020	0.51175007	0.500258867
Chiclets	-0.82031918	0.835939962	-0.06670709	0.514147274
Dots	-0.57043885	0.625011322	0.68287986	0.487777458
Dum Dums	1.60464382	-0.712637531	-0.06881335	-0.229647214
Fruit Chews	-0.64943194	-0.515343269	0.62911226	0.295017216
Fun Dip	1.64564645	-0.061325594	-0.22205535	-0.058799069
Gobstopper	0.97535806	0.006638151	0.09322099	-0.594393205
Haribo Gold Bears	-0.41418743	-0.076745232	0.97316112	0.470072508
Haribo Happy Cola	-0.96450884	0.897177682	-0.57777493	-0.279999961
Haribo Sour Bears	-0.47964306	0.144138214	0.81700174	0.481462045
Haribo Twin Snakes	-0.58554144	0.501497845	0.56435691	0.499888780
Hershey's Kisses	-0.65415252	-0.956497265	-0.15459157	-1.475048333
Hershey's Krackel	0.64433919	0.815979037	0.72869731	-0.058197261
Hershey's Milk Chocolate	0.45257976	1.365951323	0.09727415	-0.537271619
Hershey's Special Dark	0.48406856	1.259690716	0.17239789	-0.542750795
Jawbusters	0.74872541	0.777635266	-0.86953189	-0.404938915
Junior Mints	-0.56772916	-0.100807494	-0.27701048	-1.244289684
Kit Kat	0.74560410	-0.651964126	1.26789202	-0.308264629
Laffy Taffy	0.12742282	-0.323401158	0.50907401	0.902891431
Lemonhead	1.59348201	-0.612577970	-0.46974919	-0.088239458
Lifesavers big ring gummies	0.28553918	-0.405905368	0.76717408	0.968451468
Peanut butter M&M's	0.70534888	-0.535487043	-0.31945944	-0.055419751
M&M's	-0.42260375	-0.093849007	0.23580574	-1.272394704
Mike & Ike	-0.54748868	0.068855836	0.96217007	0.350296998
Milk Duds	-1.94930706	-1.088570940	-1.46793514	-0.532609515
Milky Way	-1.43541753	-0.603255019	0.91522507	-0.708957591
Milky Way Midnight	-1.60495490	-0.577210751	0.75050728	-0.825932767
Milky Way Simply Caramel	-0.81154750	-0.095588630	-0.56663709	0.057913957
Mounds	0.34136052	1.562117376	-0.17156397	-0.536908890
Mr Good Bar	1.49882941	1.178267833	-0.70645139	0.710404328
Nerds	1.05269810	-0.612484674	0.36162582	-0.677966908
Nestle Butterfinger	1.66950147	0.246674393	-0.03583585	0.547351886
Nestle Crunch	0.66634611	0.310195315	0.85793736	-0.137897493
Nik L Nip	-0.74139722	2.364290586	-0.37097983	0.877307522
Now & Later	-0.64458895	0.275284807	0.43215999	0.459054334
Payday	0.57021053	1.685803617	0.71659189	0.920899230
Peanut M&Ms	0.67649842	-0.481270608	-0.49445201	-0.017712739
Pixie Sticks	-1.00053741	-0.249857693	-0.45261662	-0.491657159
Pop Rocks	0.96236780	1.036365445	-0.40726527	-0.314719629
Red vines	-0.69083254	-0.068691578	0.66734570	0.288409391
Reese's Miniatures	1.52874138	-1.896976303	-0.34129661	0.391227548

Reese's Peanut Butter cup	1.62984822	-1.101118658	-0.10711621	0.504507095
Reese's pieces	0.71687442	-0.652294629	-0.48369702	0.001608461
Reese's stuffed with pieces	1.50741757	-0.638134777	-0.27656579	0.488051492
Ring pop	1.69809389	1.498435385	-0.64909630	0.323514628
Rolo	-1.75944281	-0.678446043	-1.06073297	-0.430807299
Root Beer Barrels	0.42962062	-0.109312144	-1.19510153	-1.555611980
Runts	0.90293644	-0.227471967	0.05500446	-0.683423916
Sixlets	-0.89038600	-0.174585492	-0.66527067	-1.454385221
Skittles original	-0.37037605	-0.800924438	1.50687697	0.245536582
Skittles wildberry	-0.46191313	-0.492027764	1.28849431	0.261464394
Nestle Smarties	-0.71703419	1.677902657	-1.00236322	-0.943766974
Smarties candy	0.89831080	-0.766691658	-0.09224917	-0.697074269
Snickers	-0.32409247	-0.999567686	0.29375023	0.519082351
Snickers Crisper	0.28905186	-0.982077142	-0.99809936	1.715122168
Sour Patch Kids	-0.44613213	-0.989653570	1.01684337	0.317967197
Sour Patch Tricksters	-0.52684967	-0.717267988	0.82427320	0.332012368
Starburst	-0.34591253	-1.030380238	1.20373631	0.352584423
Strawberry bon bons	0.76654988	-0.423292666	-0.21835972	-0.752170933
Sugar Babies	-2.27338367	0.557052968	-1.67505130	0.548087948
Sugar Daddy	-1.58729466	-0.485682924	-1.85240975	0.941435233
Super Bubble	-0.03565422	0.216125641	0.09347559	0.939439208
Swedish Fish	-0.39241423	0.662591135	0.83851208	0.624071063
Tootsie Pop	1.88703233	-0.633721587	-0.18826747	-1.043909239
Tootsie Roll Juniors	0.05449315	0.394706045	-0.68468874	-0.662902544
Tootsie Roll Midgies	-0.77589776	-0.759442617	-0.35278919	-1.510639321
Tootsie Roll Snack Bars	0.28510912	0.328392863	0.22536722	-0.875755689
Trolli Sour Bites	-0.56414212	-0.168850396	0.72703429	0.389139607
Twix	-0.49203516	-1.029755076	0.13814854	0.601927243
Twizzlers	0.17417770	-0.481177950	0.62061848	0.894755897
Warheads	1.59506307	-0.576609554	-0.45482313	-0.087796992
Welch's Fruit Snacks	-0.58742724	0.067119205	0.62144223	0.428665533
Werther's Original Caramel	0.00698052	-1.343145309	-2.51219819	-0.091968746
Whoppers	-0.46170658	0.678567795	0.16799768	-0.639281286
	PC9	PC10	PC11	
100 Grand	0.379148201	-0.154099543	0.141903843	
3 Musketeers	0.023832701	-0.155239068	-0.517927160	
One dime	-0.453988975	-0.943783623	-0.515870807	
One quarter	-0.638889823	0.095579648	-0.986719399	
Air Heads	-1.383012190	-0.573723493	-0.514453684	
Almond Joy	-0.139059851	-0.312778705	1.061148660	
Baby Ruth	1.143067977	0.264099876	-0.009085882	
Boston Baked Beans	1.396216830	-0.973188567	0.102073964	
Candy Corn	-0.173119296	-1.580718107	-0.026369468	

Caramel Apple Pops	-0.916479426	-0.201217892	0.098522113
Charleston Chew	0.265940391	-0.852594237	-0.299018234
Chewey Lemonhead Fruit Mix	-0.056700856	-0.274516517	0.133783407
Chiclets	0.600474625	0.299142845	0.263138381
Dots	-0.109595061	-0.122162631	0.085957526
Dum Dums	-0.385161810	-0.858986682	0.026635152
Fruit Chews	0.431017129	0.094469913	0.405402342
Fun Dip	-0.432729164	-0.348623806	-0.240086957
Gobstopper	0.401434380	0.160051400	0.173108273
Haribo Gold Bears	-0.021735861	0.618649969	-0.035818163
Haribo Happy Cola	0.174940628	-0.664580531	-0.210044044
Haribo Sour Bears	0.026531168	0.479623974	0.007823919
Haribo Twin Snakes	0.104620713	0.254698661	0.078430922
Hershey's Kisses	0.015418665	-0.009459574	0.108560740
Hershey's Krackel	0.589905081	-0.259828939	-0.295461506
Hershey's Milk Chocolate	-1.211270861	0.556641044	0.566669950
Hershey's Special Dark	-1.234490726	0.623522410	0.545675013
Jawbusters	1.175121068	1.211984587	0.107699541
Junior Mints	-0.125729538	0.657349544	-0.278384695
Kit Kat	0.627262392	-0.428229606	-0.052463331
Laffy Taffy	-0.694575541	-0.358253574	-0.196159662
Lemonhead	0.133629280	0.441715610	-0.165939908
Lifesavers big ring gummies	-0.856144679	0.130909678	-0.425911984
Peanut butter M&M's	0.267660757	-0.221930007	-0.095957385
M&M's	-0.712293695	0.050005905	-0.359922697
Mike & Ike	-0.220487198	-0.593471988	0.252731956
Milk Duds	-0.136911155	1.039501593	0.120288784
Milky Way	0.004765911	0.843304841	-0.327048378
Milky Way Midnight	0.046247663	-0.050365745	-0.014682753
Milky Way Simply Caramel	-1.628250067	0.337544026	1.024026651
Mounds	-1.038014053	0.444568191	0.664244259
Mr Good Bar	-0.083279402	0.319687965	0.860817575
Nerds	0.395546909	0.241504770	0.214795902
Nestle Butterfinger	-0.418486899	-0.056067346	0.931659519
Nestle Crunch	0.670459908	-0.224206267	-0.210210679
Nik L Nip	0.390203352	1.144571263	-0.293659295
Now & Later	0.340332791	0.362285847	0.182098894
Payday	1.592516886	-0.586376631	-0.517961675
Peanut M&Ms	0.463012748	0.130263439	-0.124891428
Pixie Sticks	0.506958178	-0.721036477	0.100340132
Pop Rocks	0.614743430	1.229233468	-0.196822517
Red vines	0.116010128	-0.683310675	0.459814126
Reese's Miniatures	-0.201912762	0.339292365	-0.538620165

Reese's Peanut Butter cup	-0.813354976	-0.127457497	-0.769615618
Reese's pieces	0.573552037	0.549292897	-0.190638431
Reese's stuffed with pieces	-0.924164172	-0.865127171	-0.632341326
Ring pop	-0.509522417	0.693685961	-0.801535980
Rolo	-0.716390425	0.955937401	-0.177541179
Root Beer Barrels	0.748159981	-1.361887249	0.275586696
Runts	0.490713003	-0.186267132	0.357472263
Sixlets	0.120551003	-0.694412596	0.295248736
Skittles original	-0.396609466	-0.492989386	0.235345367
Skittles wildberry	-0.329109954	-0.687411685	0.296377074
Nestle Smarties	-0.095850762	0.891841882	-0.546823295
Smarties candy	0.957784710	0.645048933	0.369028845
Snickers	1.040500931	0.639420129	-0.064039216
Snickers Crisper	1.597773064	-0.876877339	0.668926998
Sour Patch Kids	0.319739180	0.748914211	0.190355889
Sour Patch Tricksters	0.379260358	0.577472316	0.244173747
Starburst	0.178115491	0.966910650	0.055007728
Strawberry bon bons	0.831798717	-0.257396978	0.567295681
Sugar Babies	-0.204221618	-0.392704656	-0.026264253
Sugar Daddy	-0.751459910	-0.659787353	-0.269135682
Super Bubble	-0.530802993	-0.601257076	-0.099471654
Swedish Fish	-0.159360361	0.838989403	-0.260036175
Tootsie Pop	-0.721118894	0.326836317	-0.137911107
Tootsie Roll Juniors	-1.149669881	-0.284072969	-0.704615875
Tootsie Roll Midgies	0.074811904	-0.471079914	0.266947340
Tootsie Roll Snack Bars	-1.078720446	-0.723988086	1.173413519
Trolli Sour Bites	0.215399333	0.265589321	0.205354764
Twix	0.390371217	0.606070076	-0.048120393
Twizzlers	-0.729052624	-0.258947465	-0.227333151
Warheads	0.096482954	0.378764818	-0.167107014
Welch's Fruit Snacks	0.229115315	0.300496305	0.173167709
Werther's Original Caramel	0.028457521	0.448967578	-0.298604763
Whoppers	1.212121185	-1.054026246	-1.220426934

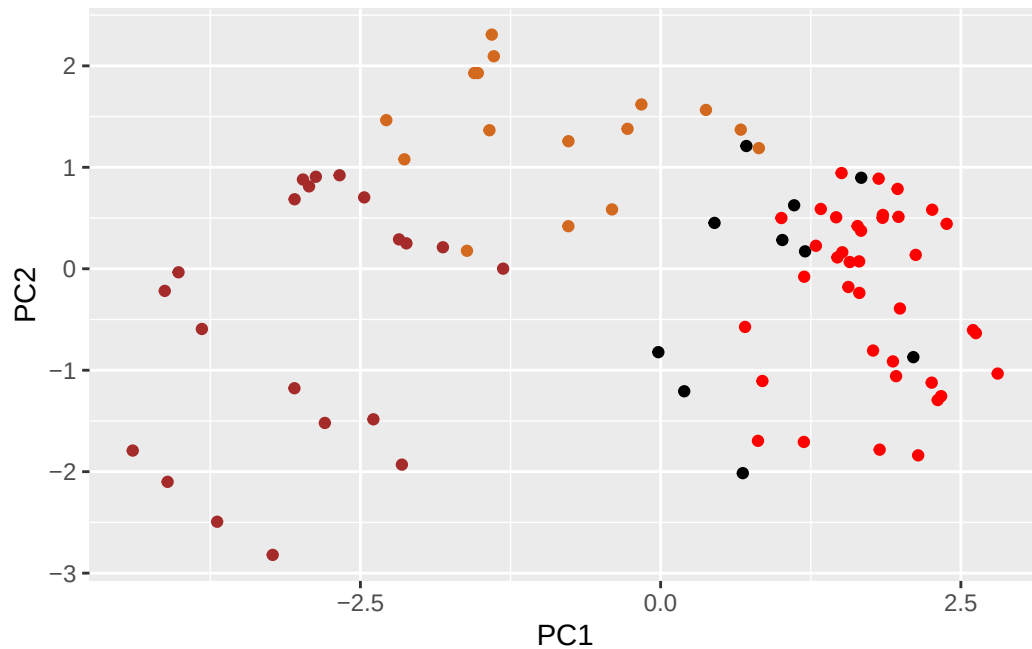
PC12

100 Grand	0.0646988318
3 Musketeers	-0.1839471662
One dime	-0.5107477914
One quarter	-0.9582719109
Air Heads	0.0442992399
Almond Joy	0.1889347102
Baby Ruth	0.5798508261
Boston Baked Beans	-0.1012981497
Candy Corn	-0.6799350975

Caramel Apple Pops	0.6962504166
Charleston Chew	0.5494017342
Chewey Lemonhead Fruit Mix	0.3105973331
Chiclets	0.5337581991
Dots	0.1503797578
Dum Dums	0.1849692597
Fruit Chews	0.1399898762
Fun Dip	0.1237990873
Gobstopper	-0.1944414618
Haribo Gold Bears	-0.2638818244
Haribo Happy Cola	-0.6879064447
Haribo Sour Bears	-0.1176800541
Haribo Twin Snakes	0.1188546962
Hershey's Kisses	0.1984492942
Hershey's Krackel	-0.0817067486
Hershey's Milk Chocolate	-0.3573702410
Hershey's Special Dark	-0.4277036644
Jawbusters	0.1337584661
Junior Mints	0.0649489483
Kit Kat	-0.3769186469
Laffy Taffy	0.3026020451
Lemonhead	0.0619011170
Lifesavers big ring gummies	-0.0228900695
Peanut butter M&M's	0.1200218111
M&M's	-0.1023879726
Mike & Ike	0.1113757406
Milk Duds	0.3400578947
Milky Way	-0.1552479006
Milky Way Midnight	0.2304632793
Milky Way Simply Caramel	-0.2533060125
Mounds	-0.1415021722
Mr Good Bar	0.0210418054
Nerds	-0.3936975444
Nestle Butterfinger	-0.3100685916
Nestle Crunch	-0.1731325150
Nik L Nip	0.4597035356
Now & Later	0.1806417362
Payday	-0.7729766006
Peanut M&Ms	0.1319358415
Pixie Sticks	-0.7378511713
Pop Rocks	-0.1936682261
Red vines	0.3438211129
Reese's Miniatures	-0.0689054242

Reese's Peanut Butter cup	-0.1005437626
Reese's pieces	-0.0005861225
Reese's stuffed with pieces	0.2335945192
Ring pop	0.0735408387
Rolo	0.0787870205
Root Beer Barrels	-0.7082327026
Runts	-0.0585635801
Sixlets	0.7459023430
Skittles original	-0.2795750540
Skittles wildberry	-0.0751177198
Nestle Smarties	0.4628839641
Smarties candy	-0.2022435492
Snickers	0.0911696493
Snickers Crisper	0.6306015108
Sour Patch Kids	-0.3186623415
Sour Patch Tricksters	-0.1383715363
Starburst	-0.5130782116
Strawberry bon bons	0.1543709171
Sugar Babies	-0.4541934821
Sugar Daddy	-0.2885121237
Super Bubble	0.6537195364
Swedish Fish	-0.2507351031
Tootsie Pop	1.2506316262
Tootsie Roll Juniors	0.5695856300
Tootsie Roll Midgies	0.4724506497
Tootsie Roll Snack Bars	-0.0373596827
Trolli Sour Bites	0.0146955266
Twix	-0.3530484324
Twizzlers	0.1981702880
Warheads	0.0702649429
Welch's Fruit Snacks	0.0727669700
Werther's Original Caramel	-0.8335356607
Whoppers	0.7201599382

```
ggplot(pca$x) +
  aes(PC1, PC2, label=row.names(pca$x)) +
  geom_point(col=my_cols)
```

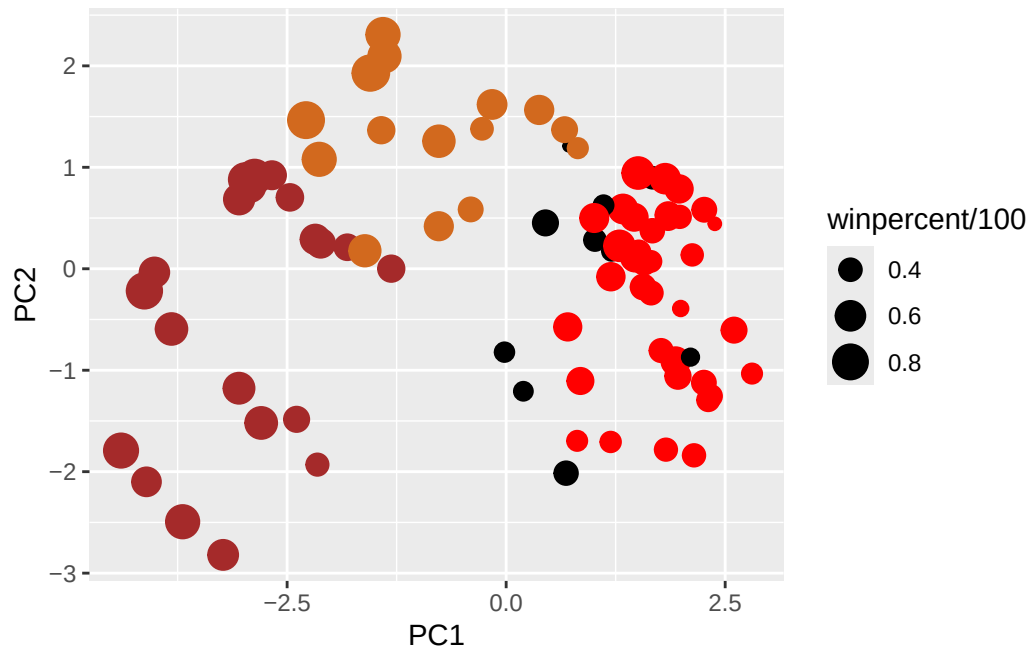


We can make a much nicer plot with the ggplot2 package.

```
# Make a new data-frame with our PCA results and candy data
my_data <- cbind(candy, pca$x[,1:3])
```

```
p <- ggplot(my_data) +
  aes(x = PC1, y = PC2,
      size = winpercent/100,
      text = rownames(my_data),
      label = rownames(my_data)) +
  geom_point(col = my_cols)
```

p

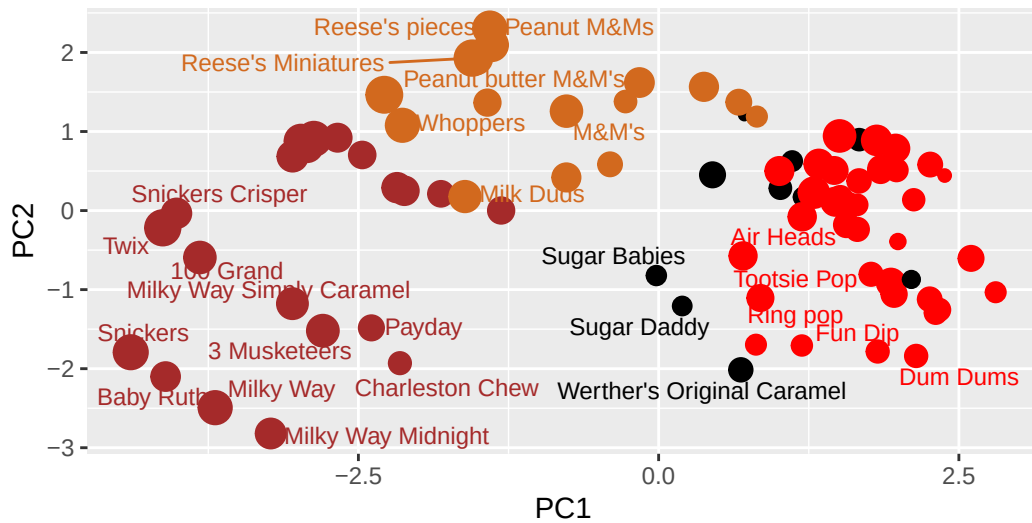


```
library(ggrepel)

p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
        subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),",
        caption="Data from 538")
```

Halloween Candy PCA Space

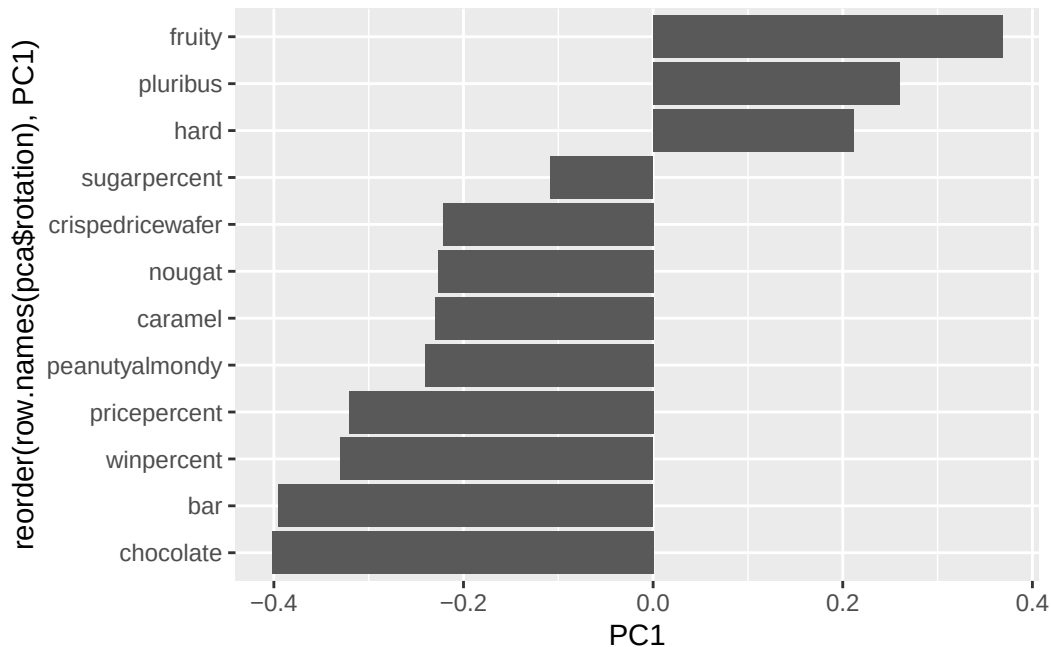
Colored by type: chocolate bar (dark brown), chocolate other (light brown),



Data from 538

The second major results figure from PCA is the so-called “loadings plot.”

```
ggplot(pca$rotation) +
  aes(PC1,
       reorder(row.names(pca$rotation), PC1)) +
  geom_col()
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

Chocolate and bar are picked up strongly as a positive contributor and fruity is picked up strongly as a negative contributor by PC1 in the positive direction. This makes sense because the same pattern is seen in the correlation plot.

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

“Winning” candies tend to have chocolate and are candy bars, but are not fruity candies. Additionally, they have decent pricings. The different analyses support this through visualization of the histograms and bar plots showing a higher win percent for chocolate candies. Moreover, a t-test was performed which validated that chocolate candies are significantly more popular. The correlation analysis showed chocolate being the candy that was positively associated with popularity, and PCA showed PC1 separating chocolate vs fruity candies with chocolate being the more popular candy. Hence, chocolate candies is more preferred and are the “winning” candies.